Dosimetry, half-life time and statistical analysis of radioactive decay of Cs-137 and Ba-137

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The radiation of a Cs-137 sample was measured at different distances. The threshold voltage associated with a Geiger-Mueller counter was determined. The responsiveness to γ- radiation was also determined to 0.2%. The half-life $T_{\frac{1}{2}} = (150.0 \pm 1.4)$s of Ba-137m was measured experimentally. A χ^2 test was conducted for a decay of uranium trioxide tile. The decay was measured with a gate time of t=10s and t=1s. It was determined that the radioactive decay follows a Poisson distribution on a five% level. The same test was executed for a Gaussian distribution which was rejected based on the 5% χ^2 test.
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1 Introduction

Radioactivity, the property of unstable atomic nuclei to spontaneously emit ionizing radiation to the environment, is a process that still plays an important role today. The effects of ionizing radiation can be very dangerous to the human body. The emitted radiation, usually in alpha, beta or gamma rays, when penetrating matter, excites the hit atoms or molecules and transfers energy. This transfer can spatially displace atoms, knock electrons out of their atomic shells, and break chemical bonds. In addition, it is important to protect oneself adequately and take precautions when working with radioactive materials because, unlike other radiation such as heat radiation, the effect is not felt immediately. However, ionizing radiation and atomic decays also have very beneficial properties, by using the radiation it is possible to destroy cancer cells and radioactivity is the basis of nuclear reactors. In this paper, the properties of radioactive decay, the half-life and some other studies on the Geiger counter are examined.

2 Fundamental theory

One can predict radioactive decays only statistically. In the case of an individual atom, it is not possible to foresee the exact time of decay. However, it is possible to calculate the probability of a decay during a period of time. Hence it is possible to determine the number of atoms N(t) remaining after a period of time if the number of atoms N_0 at $t = 0$s is known as well as the decay constant λ:

$$N(t) = N_0\, e^{\lambda t} \tag{1}$$

The half life $T_{1/2}$ is the time when half of the atoms has decayed and can be ascertained by the means of the decay constant λ:

$$\lambda = \frac{ln(2)}{T_{1/2}} \tag{2}$$

The activity of the radioactive preparation is described via $A(t) = \frac{dN}{dt}$ and was measured in the following experiment.

3 Dosimetry and radiation exposure

While conducting experiments with radiation, it is very important to monitor the radiation exposure as there is a limit to which one should be subjected. For people working with radioactive radiation there are weaker radiation limits than for the average citizen but they are closely supervised. Those limits cannot be exceeded and if so, they cannot continue working until their limit per period of time is low enough. Those limits vary with the physical condition of the regarding person.
This limit is described by the equivalent dose H and is calculated via

$$H = q \cdot D. \tag{3}$$

Whereas q scores the intensity of the radioactive radiation, it extends from q=1 for β- and γ-radiation to q=20 for α-radiation.
In this part of the experiment we examined the radiation exposure during the whole conduction. The

d/[m]	$\frac{dH}{dt}$ / [μSv/h]	H_{4h}/[μSv]
$\approx$ 0.05	6.54±0.16	26.16±0.64
1	0.250±0.054	1.00±0.22

Table 1: Equivalent dose of the radioactive preparation depending on the distance between the preparation and the measuring tool per hour and 4 hours.

equivalent dose H is listed with the according distance d to the probe in tab. 1.
The period of time until reaching 5mSv being 1m apart from the radioactive preparation amounts to t=(2.00±0.43)·10^4 h. The relative ratio between the period of time reaching the 5mSv and the 4h executing the experiment numbers to p=(0.0200±0.0043)%. Then the radiation exposure was determined and is given in tab. 2.

$H_{3.28h}$/[μSv]	H_{1a}/[mSv]
0.375± 0.092	1.00 ± 0.25

Table 2: Radiation exposure during the experiment plus the one for a whole year in the laboratory.

It is visible, that the radiation exposure during the experiment and even extrapolated for a year does not overstep the radiation limit of H=1mSv for an average person.

4 Activity of Cs-137

The activity A(t) is described by

$$A(t) = \frac{dN}{dt} = A_0 e^{-\lambda t} \tag{4}$$

It describes the decays per period of time. Knowing the half-time and the date of manufacture, we can calculate the activity of the Cs-137 probe of today:

Date of manufacture	21.08.1978
Date of execution	01.04.2022
Activity A_0	3.7 $\cdot 10^6 Bq$
Half-life time	30.2a

Table 3: Given data to calculate the activity of Cs-137

With the given data from tab. 3 and eq.[4] the activity is calculated to

$$A = (1.3607 \pm 0.0010)\, MBq$$

5 Measurements with a Geiger-Müller counter

5.1 Function of a Geiger-Müller counter

The Geiger-Müller counter measures power surges which are triggered by γ-radiation ionising gas atoms. The Geiger-Müller counter is a capacitor with a positive charged wire in the middle surrounded by a negative charged cylinder. The electric field is stronger in the middle because of the cylindric-structure:

$$E = \frac{1}{r}\frac{U}{ln(r_1/r_2)} \tag{5}$$

The counter is filled with argon and alcohol vapour. The γ-radiation ionises the gas. Thereupon free electrons create an ionisation avalanche near the positive charged wire and the following power surge creates a voltage surge at a resistor connected in between. This voltage surge is transferred to another capacitor which is connected to an amplifier. Since the ionised atoms move slower to the negative charged cylinder than the free electrons to the positive charged wire because they weigh more, the electric field between those ionised atoms and the positive charged wire is decreased as the ionised atoms shield the negative charged cylinder. The ionisation avalanche stops. When the ionised atoms reach the negatively charged cylinder, a new ionisation avalanche can be caused.

5.2 Characteristics of a Geiger-Müller counter

The activity of a radioactive probe is measured through the voltage surge created at a resistor according to the Ohm-law. This power surge depends on the applied voltage. If this voltage is too small, the electric field in the Geiger-Müller counter may be too small to create an ionisation avalanche and no ionisation and subsequently no activity of the probe can be measured. The minimum voltage at which a current surge can be detected is called threshold voltage.
To determine the threshold voltage, we increased the applied voltage until we could detect voltage pulses. The threshold voltage amounts U=(516±4)V according to fig. 1. The error results from the uncertainty of adjusting the applied voltage.
To measure enough voltage impulses in the following parts of the experiment, we decided to use an input voltage of U=560V.

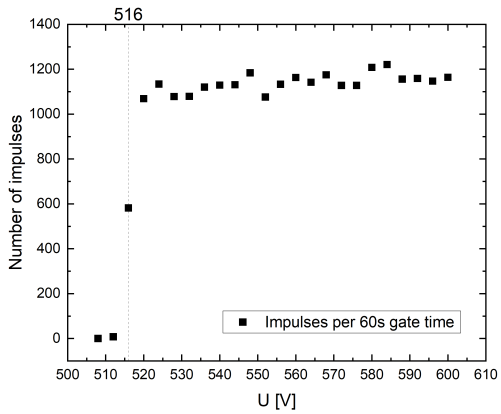

Figure 1: The measured impulses are portrayed in reference to the input voltage. Voltage surges are detected above a threshold voltage of U=(516±4)V.

5.3 Zero effect

In order to determine the actual radiation activity of materials, it is important to know beforehand that a so-called zero effect exists. This effect means that although there are no radiating samples in the vicinity, the Geiger-Mueller counter still detects radiation. The reason for these measurements is cosmic radiation, as well as partially occurring radioactive isotopes. To measure to the zero effect, the measuring device is shielded with lead walls and the detected radiation is recorded. Subsequently, this measurement is repeated at the workplace without shielding. Both procedures are carried out over a period of $t = 5$min with a voltage of $U_{tube} = 560$V. The resulting values are shown in Table 4.

counter location	null effect [min^{-1}]
working space	38.2
inside shielding	19.0

Table 4: Measuring the null effect for a Geiger counter at different locations at the workplace

As expected, the detected radiation decreases strongly inside the lead shielding. Proportionally, the value of radiation outside to the inside decreased by 49.74%.

5.4 Measuring the half- life of Ba-137m

The Cs-137 used in the experiments can enter two states by β- radiation. One is the stable Ba-137, but the other is the excited state of Ba-137m. By subsequent radiation of γ-rays, the Ba-137m also reaches its ground state. In order to obtain this excited barium, it is now possible to separate the Ba-137m from the Cs-137 due to the different solubilities. Subsequently, the half-life of the dissolved sample was measured. For this purpose the Ba-137m was placed in front of the Leybold Geiger counter and the number of measured decays was noted during a gate time of $t = 10$s.

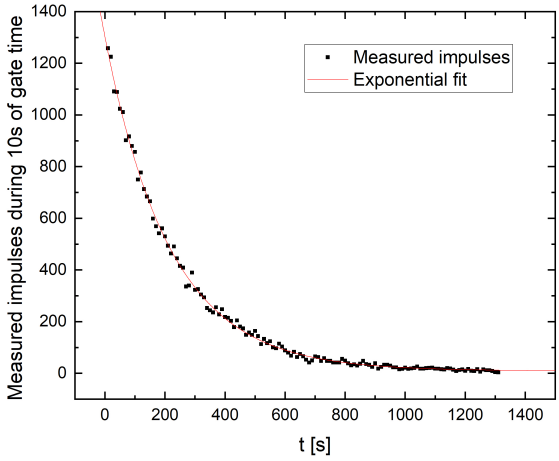

Figure 2: Decay rate of a Ba-137m sample fitted with an exponential function. The exponent of the fit allows us to calculate the half life of Ba-137m as $T_{\frac{1}{2}} = (150.0 \pm 1.4)s$.

The resulting data is shown in fig. 2.
In order to calculate the half-life with the measured decays, an exponential curve was fitted to the data in fig. 2 which provides us $\lambda = 0.004620 \pm 0.000044$. With λ and formula 2 the half-life of Ba- 137m can be determined to $T_{\frac{1}{2}} = (150.0 \pm 1.4)$s. This experimentally determined value is not consistent within its error with the theoretical value of $T_{1/2}$=153.12s [3].

5.5 Responsiveness of the Geiger-counter

In this part of the experiment, the responsiveness of the Geiger counter is calculated. In reality, not every incoming γ- particle interacts with the gas inside the tube. The ratio of the incoming and detected radiation is called the responsiveness of the counter with respect to gamma radiation. To obtain this value for the Leybold counter used here, the Cs-137 sample is placed at a distance $d = (0.80 \pm 0.01)$m in front of the counter. Then the number of detected impulses is measured in a time interval of $t = 10$s. The responsiveness $\gamma = (0.2030 \pm 0.0010)\%$ was determined with

the formula

$$\gamma = \frac{N}{A\,t\,\frac{F}{4\,\pi\,d^2}} \tag{6}$$

Here F corresponds to the sensitive area of the counting tube, N to the measured pulse rate in $t = 10$s and A is the calculated activity in section 4.

6 Analysis of radioactive decays

6.1 Poisson and Gaussian distribution

Equation 1 describes the expected value of the number of non-decayed atoms after a period of time t. However, the radioactive decay is to be described stochastically and the number of decays scatters around the expected value. To get a better understanding of this behavior, it is useful to take a closer look at the distribution:
The probability of k atoms to decay during the gate time is given by the Poisson distribution with

$$P(k) = \frac{\mu^k}{k!} e^{-\mu} \tag{7}$$

Whereas μ is the mean decay number detected.
For big mean $\mu = p_{decay} \cdot N_{atoms}$ the Poisson distribution transitions into a Gaussian distribution. The probability of k decays during a gate time is then given by

$$P(k) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(k-\mu)^2}{2\sigma}} \tag{8}$$

The standard derviation is given by $\sigma = \sqrt{\mu}$.

6.2 Poisson and Gaussian distribution with gate time t=10s

For this part of the experiment a radioactive orange tile was put in a plumb box with the counting tube Leybold to suppress the background radiation. The number of decays during a gate time of t=10s was measured for 1000 gate times. To ascertain if the detected number of impulses follows a Gaussian or Poisson distribution, we conduct a χ^2-test. At first the number of detected impulses for a certain decay number per gate time was listed and the theoretical expected values were determined for Gaussian and Poisson distribution. Then the number of classes was compiled. The number of detected decays for each class is portrayed in fig. 3. The mean μ accounts to $\mu = 7.12 \pm 0.37$. The mean is pretty high so we examine a χ^2-test for a Poisson and Gaussian distribution. χ^2 indicates the deviation from a distribution. The smaller the value of χ^2 the more likely that the data follows the reference distribution. The calculated values are listed in tab. 5.

Null hypotesis	χ^2	$\chi^2_{5\%,max}$
Poisson distribution	12.70	23.03
Gaussian distribution	23.91	21.03

Table 5: χ^2-values are listed to ascertain if the number decays of a radioactive tile during a gate time of t=10s follows a Poisson and Gaussian distribution.

The χ^2 value with a significance level indicates that if the calculated χ^2 is higher, the deviation is too big and the distribution does not follow the examined one. The $\chi^2_{5\%,max}$ indicates the threshold of χ^2. If we calculate a χ^2 higher than the one of the significance level of 5% and conclude that it does not follow a Poisson or Gaussian distribution, there is a 5% chance left that it actually does.
As we can see the calculated χ^2 for Poisson distribution is smaller than the value of a significance level of 5%. For the Gaussian distribution, the calculated χ^2 is bigger than the one with a significance level of 5% and this hypothesis needs to be rejected. In conclusion, this distribution follows a Poisson distribution and not a Gaussian.

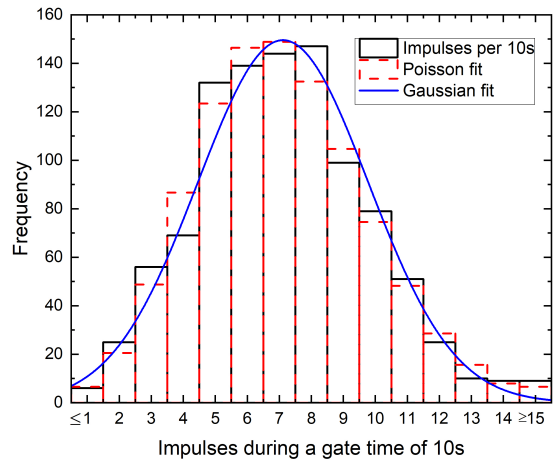

Figure 3: The measured frequency of a number of decays during a gate time of t=10s, as well as the Poisson and Gaussian distribution, is depicted.

To stress this assumption, our measured data is plotted with its cumulated relative frequency along the number of measured decays during the gate time in fig. 4. Since the probabilities do not follow the linear fit and the residuum is not by chance, the data does not likely follow a Gaussian distribution. The ascertained mean based on fig. 4 is listed in tab. 6.

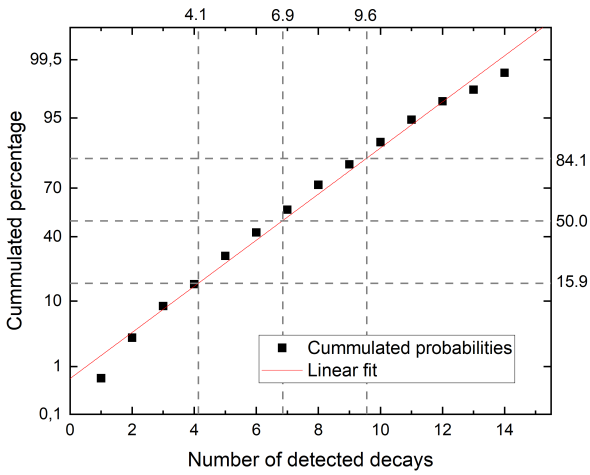

Figure 4: Cumulated relative frequency mapped onto measured decays during a gate time if t=10s.

Calculated mean	7.12±0.37
Ascertained mean	6.9±2.8

Table 6: Listed are the out of the data calculated mean and the one ascertained graphically by the help on fig. 4.

The standard deviation ascertained by the help of fig. 4 is determined by the help of the Gaussian bell and its probability that 68.2% of the data lies within this range.

6.3 Poisson and Gaussian distribution with a gate time of t=1s

In this part we also used the radioactive orange tile with the counting tube Leybold in a plumb box but measured now the number of decays during a gate time of 1s. The mean numbers to $\mu = 0.7 \pm 1.1$.

Null hypotesis	χ^2	$\chi^2_{5\%,max}$
Poisson distribution	5.14	7.81
Gaussian distribution	175.03	5.99

Table 7: χ^2-values are listed to ascertain wether the number decays of a radioactive tile during a gate time of t=1s follows a Poisson and Gaussian distribution.

This mean is pretty small, smaller than the one in section 6.2. Hence, the distribution will probably not follow a Gaussian distribution as Gauss approaches Poisson only for big mean μ. The χ^2 values are listed in tab. 7.
As assumed, our data does not follow a Gaussian but a Poisson distribution. The distribution of the number of impulses during a gate time of t=1s is depicted in fig 5.

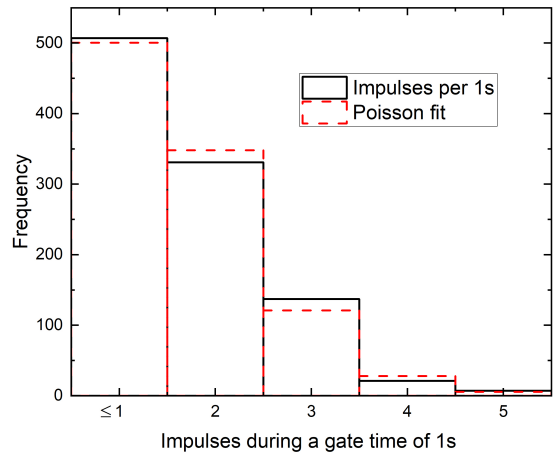

Figure 5: The measured frequency of a number of decays during a gate time of t=1s, as well as the Poisson distribution, is depicted.

7 Results

For radiation exposure we measured the dose rate $\frac{dH}{dt}$ in a distance of d=0.05 as $\frac{dH}{dt} = (6.54 \pm 0.16)\frac{\mu Sv}{h}$ and calculated the dose for 4 hours during the conducted experiment as H_{4h}=$(26.16 \pm 0.64)\mu$Sv. In one meter distance, the dosis rate numbers to $\frac{dH}{dt} = (0.250 \pm 0.054)\frac{\mu Sv}{h}$ and the dose for 4 hours accounts to $H_{4h} = 1.00 \pm 0.22)\mu$Sv. The relative ratio between reaching a limit of 5mSv and the dose rate in 4 hours aggregate to p=(0.0200±0.0043)%.
Then the activity of a Cs-137 probe was calculated to A=(1.3607±0.0010)MBq. With this activity we determined the responsiveness of a Geiger-Müller counter to $\gamma = (0.2030 \pm 0.0010)\%$.
The threshold voltage of the used Geiger-Müller counter was ascertained to $U_{thres} = (516 \pm 4)V$. For the null effect the background radiation was measured by the help of the Geiger-Müller counter: at first at the working space to 38.2 counts per minute and inside a plumb shielding to 19.0 counts per minute. The deviation is 49.74%.
Then the half-life-time of Ba- 137m was determined to $T_{1/2} = (150.0 \pm 1.4)$s.
Lastly, the number of decays of a radioactive orange tile was ascertained for t=10s and t=1s. The mean of decays to the regarding gate time was calculated to $\mu_{1s} = (0.7 \pm 1.1)$, $\mu_{10s,calc.} = (7.12 \pm 0.37)$ and

$\mu_{10s,graph} = (6.9 \pm 2.8)$. For both gate times the χ^2 values acknowledged that the decay of a Ba- 137m follows Poisson but rejects Gauss distribution.

References

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[3] Half life of Ba-137m, https://www.internetchemie.info/isotop.php?Kern=Ba-137m, last visited April 8th, 2022, 11:48 am

[4] Physikalisches Grundpraktikum, Universität Würzburg, "Begriffsbestimmungen Strahlenschutz", 2022